

Linear Regression

Prediction, loss, and feature representation

Linear regression predicts a number from input features. The model defines how features produce predictions, the loss defines a good fit, and the feature representation determines which relationships the model can express.

Learning goals. By the end of this lecture, you should be able to

1. identify a regression problem and interpret the weight and bias in a linear model;
2. explain why signed error is unsuitable as a loss and compute mean squared error;
3. explain how a feature map lets a linear model represent a curved relationship.

1 Begin with the prediction problem

Consider predicting a student's GPA from a feature such as study time. A scatterplot might show GPA increasing with study time or decreasing with gaming time. Linear regression represents either trend with a line.

Classification and regression. Classification predicts a category, such as whether an image contains a cat. Regression predicts a number, such as a student's GPA. The output type determines the task. Here the target is numerical, so this is a regression problem.

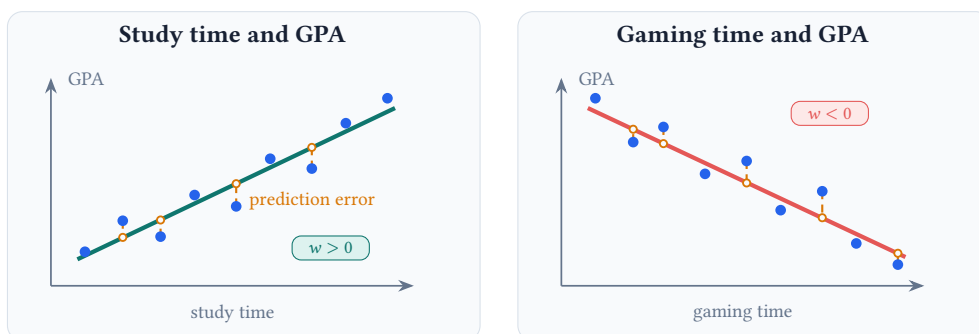


Figure 1. Two regression examples with synthetic data. For each input, the fitted line gives a prediction $\hat{y}_i$. Each vertical segment connects the observed value y_i to its prediction and shows the error.

For observations $\mathcal{D} = \{(x_i, y_i)\}_{i=1}^m$, the one-feature model predicts

$$\hat{y}_i = wx_i + b. \quad (1)$$

The input x_i is a *feature*, y_i is the observed *target*, and $\hat{y}_i$ is the model's prediction. The two learned coefficients play different roles:

Quantity	Role	Effect on the prediction
w	weight or slope	Changes how much $\hat{y}$ moves when x increases by one unit.
b	bias or intercept	Shifts every prediction by the same amount.
$\hat{y}_i$	predicted target	Gives the model's estimate for example i .

Table 1. The components of a one-feature linear model.

With several features, the model becomes $\hat{y}_i = \mathbf{w}^\top \mathbf{x}_i + b$. It assumes that the chosen features contain useful information about the target and that the same coefficients apply across examples. Training chooses the coefficients that make the predictions close to the observed targets.

2 Define what counts as error

A model needs a numerical definition of a good prediction. For example i , define the residual as

$$e_i = y_i - \hat{y}_i. \quad (2)$$

Why signed error fails

A common first reaction is to use the signed difference $\ell_i = y_i - \hat{y}_i$ as the loss. When these signed errors are averaged over the dataset, positive and negative values can cancel. More importantly, minimizing this loss rewards arbitrarily large predictions: increasing $\hat{y}_i$ drives ℓ_i toward $-\infty$ even as the prediction gets worse. This is a simple form of reward hacking: the model improves the score without improving the prediction. Absolute and squared loss close the loophole by assigning every error a nonnegative penalty.

Two common ways to turn a residual into a per-example loss $\ell(y_i, \hat{y}_i)$ are

Loss	Definition	Behavior
Absolute loss	$ y_i - \hat{y}_i $	The penalty grows in direct proportion to the size of the error.
Squared loss	$(y_i - \hat{y}_i)^2$	The penalty grows quadratically, so larger errors receive more weight.

Table 2. Two ways to score the error on one example. This lecture uses squared loss.

For squared loss, the training loss is the average squared error across the dataset:

$$L(w, b) = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2 = \frac{1}{n} \sum_{i=1}^n (y_i - (w x_i + b))^2. \quad (3)$$

This quantity is the *mean squared error* (MSE).

Example	Target y_i	Prediction $\hat{y}_i$	Squared error
1	7	10	$(7 - 10)^2 = 9$
2	5	5	$(5 - 5)^2 = 0$
3	7	5	$(7 - 5)^2 = 4$

Table 3. A three-example calculation gives $\text{MSE} = (9 + 0 + 4)/3 = 13/3$.

A lower MSE means that the squared errors are smaller on average for this dataset.

3 Representation sets what the model can express

To model a curved relationship, we can transform the original input x into polynomial features before fitting the model. For degree d , define

$$\phi(x) = \begin{bmatrix} 1 & x & x^2 & \dots & x^d \end{bmatrix}^\top, \quad \hat{y} = \theta^\top \phi(x). \quad (4)$$

The prediction can now curve with x . The parameters still enter linearly: in $\theta_0 + \theta_1 x + \theta_2 x^2$, each learned coefficient appears only to the first power.

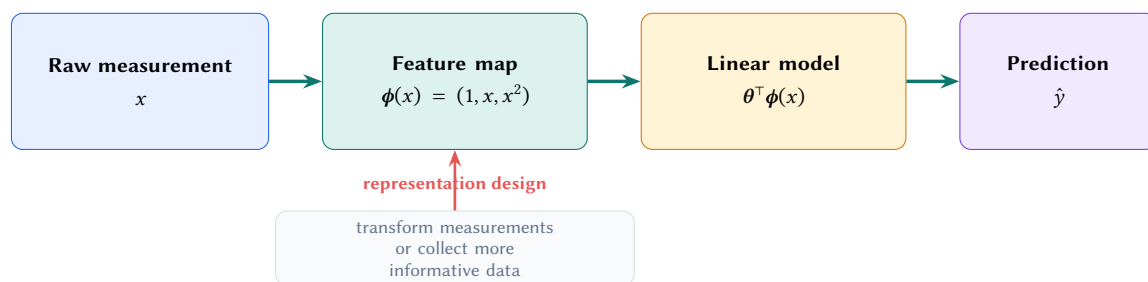


Figure 2. A model can use only the features it receives. Transforming measurements or collecting more informative data can expose relationships that a linear predictor can learn.

Residuals can show when the current features are inadequate. A systematic curve in the residuals means that the current features leave predictable structure unexplained. Adjusting the coefficients alone cannot fit a pattern the representation cannot express. Change the feature map or collect better measurements.

Lecture summary

Linear regression predicts a number from a weighted combination of features. The loss defines a good fit, and the feature representation determines which relationships the model can learn.

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